Zohaib Aftab, Ph.D.

AI Architecture Specialist | Generative AI & Enterprise Solutions
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Summary

AI architect and data scientist with 10+ years of experience across industry and research. Designs and deploys enterprise AI, machine learning, and generative AI solutions on Azure. Combines hands-on engineering with reference architecture design, cross-functional technical leadership, and MLOps/LLMOps practices. Ph.D. in robotics; bilingual in French and English.

Experience

Sept 2024 – present

Specialist - AI
Synechron

AI solution architecture and delivery for US Bank wealth and asset management.

- Led a production generative AI assistant integrating SQL, Swagger API references, Confluence, and SharePoint through retrieval-augmented generation (RAG). Adopted by 600 users, it reduced API discovery time by 90% and saved over $1M in annual operating costs.
- Designed Python microservices with async FastAPI, SQLAlchemy, and connection pooling; deployed on Azure Kubernetes Service (AKS) with horizontal pod autoscaling and request tracing.
- Architected a ServiceNow incident resolution platform with defined service boundaries, reusable interfaces, and validation schemas, enabling engineering teams to integrate their data sources.
- Delivered LangGraph multi-agent workflows across Salesforce, SharePoint, and Confluence. Combined hybrid search with source-grounded answers and evaluated accuracy, groundedness, recall, and latency.
- Architected a document intelligence platform with Azure Document Intelligence and LLMs, using configuration-driven extraction schemas to support new business lines without code changes.
- Establish reference architectures and testing playbooks, mentor engineers, and support pre-sales solution design and technical proposals. Responsibilities include system design and data governance.

Oct 2023 – Sept 2024

Senior Data Scientist
B-yond – Montreal, Canada

- Built a RAG application over 50,000+ pages of technical documentation and PostgreSQL data, achieving over 90% precision and recall.
- Engineered Apache Spark and Databricks pipelines on Azure; developed time-series models for Wi-Fi quality-of-service trends and LDA/BERTopic models for Orange's JIRA issue resolution.
- Collaborated with architecture and frontend teams to deploy AI applications, using CI/CD pipelines and MLflow model tracking for reliable delivery.

July 2018 – June 2023 **Research Data Scientist**
National Center of Robotics and Automation – Pakistan

- Directed cross-functional teams on government projects with budgets exceeding $200,000, delivering 2 commercial AI products and securing 3 patents.
- Built pipelines for 8 million+ data points spanning 11+ hours of sequential data; led temporal CNN development in TensorFlow for human activity recognition.

Technical skills

AI architecture & integration: Solution design, reference architectures, enterprise integration, Python microservices, data governance, SDLC, Scrum

Machine learning & generative AI: Deep learning (CNNs), LLMs, RAG, multi-agent systems, LangGraph, LangChain, MCP, TensorFlow, scikit-learn, Azure Document Intelligence

Cloud & data platforms: Azure, AKS, Docker, Kubernetes, FastAPI, SQLAlchemy, SQL, PostgreSQL, Apache Spark, Databricks

MLOps / LLMOps & observability: CI/CD, GitHub Actions, GitLab pipelines, MLflow, evaluation pipelines, Langfuse, LangSmith, Lumenova, request tracing

Certifications

2026 **Azure AI Apps and Agents Developer Associate**
Microsoft | Proctored exam (AI-103)

2026 **Claude Certified Architect - Foundations**
Anthropic | Proctored exam (CCA-F)

Education

2012 **Université de Lyon**, Ph.D. in Robotics – France

- Humanoid balance recovery using constrained optimization, quadratic programming, and model predictive control.

2009 **École Centrale Nantes**, Master's in Automation – France

Mentoring & writing

Mentor for a course with UT Austin covering Python, machine learning, and agentic engineering.

Transforming Telco Troubleshooting: Our Journey Building TelcoGPT with RAG